

Day 8 — Climate Change: What’s Happening and Why

Module: Indic Ecology · **Band:** Middle · **Time:** 45 min

Learning objective

Students explain — in their own words — what climate change is, name three observable effects, distinguish *climate* from *weather*, and articulate what the Indic frameworks they’ve studied can (and cannot) add to a modern response.

Materials

- Whiteboard
- A simple graph (printed or projected): global average temperature since 1850 (the “hockey stick”). Source: IPCC AR6 Working Group I.
- Optional: a 2-minute video clip from the IPCC’s outreach material or a reliable news source on a recent extreme-weather event
- Notebook

Vocabulary

- *climate* — the long-term average of weather in a region; measured over decades.
- *weather* — what’s happening on a particular day or week.
- *greenhouse gas (GHG)* — gases (CO₂, methane, N₂O, water vapour) that trap heat in the atmosphere.
- *carbon dioxide (CO₂)* — the main long-lived greenhouse gas humans emit.
- *anthropogenic* — caused by human activity.
- *IPCC* — Intergovernmental Panel on Climate Change; the UN body that compiles climate science.
- *SDG* — Sustainable Development Goals (UN, 2015). 17 goals; SDG 13 is “Climate Action.”

Lesson flow

Warm-up (5 min)

- Quick poll, hands up: “Who has heard the phrase ‘climate change’? Who knows what it means in one sentence?”
- Take 3 student definitions on the board, without correcting yet.

Core (25 min)

1. **Climate vs. weather — the distinction.**
 - **Weather:** “It rained yesterday.”
 - **Climate:** “The June monsoon now arrives 7 days later than it did 50 years ago.”

Weather is local + short. Climate is large + long. Climate change means the *long average* is shifting.

2. **What’s happening (the numbers, simply).** Show the IPCC AR6 graph.

- Global average temperature has risen ~1.1°C above the 1850–1900 baseline. (IPCC AR6, 2023.)
- 2023 and 2024 were the two hottest years on record by a clear margin.
- The IPCC’s “Paris-agreement” target is to keep warming below 1.5°C; we are very close to that line.

3. **Why it’s happening.** Briefly:

- CO₂ in the atmosphere has risen from ~280 parts per million (ppm) in the 1800s to ~420 ppm today.
- That CO₂ comes mainly from burning fossil fuels (coal, oil, gas) for energy and transport, from deforestation, and from cement production.
- More CO₂ → atmosphere traps more heat → average temperature rises.

We are not claiming this is the only cause; we are claiming this is the cause that science has identified as dominant since ~1950. The IPCC reports are the authoritative source.

4. **Three observable effects (pick the ones most local to your students):**

Effect	Where students might already see it
Heatwaves and longer hot seasons	Indian summer is hotter and longer than 30 years ago
Monsoon disruption	More floods or droughts; less predictable arrival
Glacier retreat	Himalayan glaciers feed Gaṅgā, Yamunā, Brahmaputra — they are shrinking

5. **Where the Indic frameworks help — and where they don’t.**

Help:

- The *yajña* cycle (Day 5) is a direct mental model of “what you take, you must give back.” Climate change is what happens when this is ignored.
- The *iṣopaniṣad* mantra (Day 7) — “*use, don’t grab*” — is the ethic of a low-carbon civilisation in five words.
- The *Prthivī Sūkta* (Day 6) — “*may what I dig grow back*” — was always the sustainability rule.
- The *vasudhaiva kuṭumbakam* ethic (Day 1) reframes climate as a *family* problem, not a national one.

Don’t help (or don’t help by themselves):

- Old frameworks don’t tell us what 1.5°C means.
- They don’t tell us how to design a solar grid, a heat-resistant rice variety, or a more efficient kiln.

- They don't substitute for measurement, monitoring, policy, and engineering.

Honest summary: the Indic frameworks give us the *attitude* and the *language* of restraint. Modern science gives us the *measurements* and the *technology*. The climate crisis needs both.

6. **What India is doing.** Brief mention:

- India has committed (Paris pledges + COP26) to reach net-zero emissions by 2070.
- National solar capacity has grown ~30× since 2010.
- National Action Plan on Climate Change has 8 missions (solar, water, energy efficiency, sustainable agriculture, Himalayan ecosystem, sustainable habitat, green India, knowledge).

This is not a politics lesson; this is just “what’s currently happening.”

Activity (10 min) — Pair brainstorm: pitch the principle

- Each pair picks ONE of the four Indic ideas they’ve studied:
 1. *vasudhaiva kuṭumbakam* (one family)
 2. *yajña* cycle (give back)
 3. *Pr̥thivī sūkta* (Earth as mother)
 4. *Iṣopaniṣad* mantra (use, don’t grab)
- They write 3 sentences: “How does this idea help us think about climate change? What does it NOT solve?”
- This is the precursor to the project pitch on Day 10. Pairs should pick whichever principle they want to anchor their project around.

Wrap-up (5 min)

- **Project plan due tomorrow (Day 9 morning).** Each pair must submit a paragraph stating:
 1. Which Indic principle they will anchor on.
 2. What local observation they will make on Day 9’s walk.
 3. What format (poster / audio / action plan) their pitch will take.
- One-sentence exit ticket: “The hardest thing about taking climate change seriously is _____. ” (Honest answers welcome.)

Student-facing content

Climate change in three sentences.

The Earth’s average temperature has risen about 1.1°C since the 1800s. The cause, since about 1950, is mostly humans burning fossil fuels and cutting forests, which puts CO₂ in the atmosphere and traps heat. The result is hotter summers, disrupted monsoons, retreating glaciers, and rising seas.

What the Indic frameworks add:

- *Yajña* cycle: what you take, you must give back.
- *Iṣopaniṣad*: use, don't grab.
- *Prthivī Sūkta*: may what I dig grow back.
- *Vasudhaiva kuṭumbakam*: the planet is one family — every nation has to act together.

What modern science and engineering add:

- The measurements (we know exactly how much CO₂ is up there).
- The targets (1.5°C, net-zero, ppm).
- The technologies (solar, wind, electric transport, efficient buildings).

Neither side is complete without the other.

Homework

- **Project plan paragraph due tomorrow morning.** See assessments/project-brief.md.
- Also: pick one news headline from the last week that mentions climate change, weather extremes, or pollution. Bring a one-sentence summary.

Differentiation

- **One tier up:** Read the IPCC AR6 Synthesis Report's "Summary for Policymakers" — first 5 pages only (it's accessible online). Bring one fact tomorrow.
- **One tier down:** Focus on one observable effect — the one most visible in your region (heatwaves, floods, monsoon shift). One effect, well understood, is enough.

Teacher notes

- Climate change is a politically loaded topic in some classrooms. Stick to the IPCC numbers; they are the consensus scientific source. Avoid the activist framing on either side.
- Some students or parents may dispute the science. The classroom position is: "the IPCC is what we are teaching. If you disagree, the place to argue with the science is with peer-reviewed responses to the IPCC, not in this classroom."
- Avoid catastrophising. Students at this age can become genuinely anxious about climate news. Frame it as: *this is real, this is serious, this is also being worked on by millions of people including engineers, farmers, and you.*
- Avoid the opposite — false hope. Don't say "if we all just plant trees we'll be fine." Plant trees AND change energy systems AND change consumption AND change policy. The Indic frame supplies one thread of this; not the whole rope.

Citation(s) used in this lesson

- **IPCC AR6 Synthesis Report (2023)** — for global temperature figures, the 1.5°C target, and CO₂ concentrations.
- **UN Sustainable Development Goals (2015)** — for the SDG 13 reference.
- **National Action Plan on Climate Change** (Government of India, 2008, with 2022 updates) — for the 8 missions framework.

These are all open / government sources; full references in `sources.md`.